Proof-of-Concept Labs: Early Network Practice

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Summary

These five Packet Tracer files are practice labs from the early stages of my learning and development as a network engineer and designer. Each one is a proof of concept: after learning a new technology, I built a small network to apply it, see how it behaves and practice designing around it. They are not finished production designs, and I treat them that way.

Taken together they show a steady progression:

- **Fundamentals first:** subnetting, VLANs, trunking, SSH-only management and switch hardening.
- **Routing practice:** EIGRP, OSPF with multiple areas, eBGP, and redistribution between protocols.
- **Overlay and redundancy:** GRE tunnels in hub-and-spoke layouts, HSRP gateway pairs, EtherChannel and DHCP on routers.
- **Security basics:** an ASA firewall with NAT and web access rules, and a honeypot server in a segmented design.

The later Three-Tier Campus and CollapseCore builds are where these ideas are combined at larger scale and then hardened.

Labs at a glance

Lab	Devices	Links	Main concepts practiced
1. OSPF-EIGRP / EIGRP-BGP campus	25	32	Multi-area OSPF, EIGRP, mutual redistribution, eBGP, GRE hub-and-spoke, HSRP, EtherChannel
2. Intranetwork V2.0 campus	67	72	Lab 1 concepts scaled up, plus ASA firewall, NAT, web ACLs, STP root priorities, 12 IP phones

Lab	Devices	Links	Main concepts practiced
3. Network Data Infrastructure LAN	23	20	EIGRP with GRE, router-based DHCP, trunk VLAN lists, SSH-only access
4. Project Network	22	20	Two EIGRP processes, GRE, ASA, public router and honeypot server, server VLAN
5. Completed final	9	9	Two mirrored sites, EIGRP, GRE, static defaults, Layer 3 switches, device hardening

The files were saved in different Packet Tracer versions (7.2, 8.2 and 9.0), which fits labs built at different points along the learning path.

Lab 1: OSPF-EIGRP and EIGRP-BGP campus network

Goal: practice connecting routing domains: OSPF and EIGRP together, then EIGRP and BGP at the edge.

- **Build:** 25 devices: 6 routers, 2 Layer 3 switches, 6 switches, 6 servers (DNS, DHCP, web, TFTP and others), a PC, a laptop and a cloud.
- **OSPF:** areas 0, 1 and 2, with a hub router connecting them.
- **Redistribution:** the hub redistributes OSPF into EIGRP 12 and EIGRP into OSPF.
- **eBGP:** the edge router (AS 6500) peers with a provider router (AS 6501) and redistributes between BGP and EIGRP.
- **GRE:** hub-and-spoke tunnels from the hub to the server router and to both branch routers, plus a tunnel to the edge.
- **Redundancy:** an HSRP pair of branch routers, with DHCP pools (users, voice, printer, VIP) on one of them; two Layer 3 switches joined by an EtherChannel.
- **Management:** SSH v2 with local login on vty lines, console and vty logging synchronous.

Learning notes: this lab centers on mutual redistribution and how it affects the routing table. Small addressing slips (such as a 128.138 tunnel address and a mistyped domain name on the access switches) were the kind of thing I learned to catch by validating every endpoint.

Lab 2: Intranetwork V2.0 campus

Goal: scale the Lab 1 design to a full campus and add a perimeter firewall.

- **Build:** 67 devices and 72 links: 5 routers, 2 Layer 3 switches, 7 switches, an ASA 5506-X, 6 servers, 26 laptops, 12 IP phones and 3 printers.
- **Routing:** OSPF areas 0, 1 and 2 with EIGRP 12, redistribution on the hub and on the ASA, and `passive-interface default` on the hub and branch routers so adjacencies form only where intended.
- **GRE:** hub-and-spoke tunnels over serial links to the server router and both branches, and a tunnel to the edge.
- **Redundancy:** HSRP between the two branch routers with DHCP pools for users, voice, printer and VIP VLANs.
- **Spanning tree:** the two Layer 3 switches are given explicit root priorities (24576 and 28672).
- **Firewall:** the ASA has inside and outside interfaces, dynamic NAT to the outside interface, and access lists that permit HTTP and HTTPS to the web server.
- **VLANs:** user, voice, printer, VIP, native (90) and management (999) VLANs carried on trunks with allowed lists.

Learning notes: the largest lab, with a firewall and enough endpoints (including voice) to feel like a real campus.

Lab 3: Network Data Infrastructure LAN

Goal: a compact hub-and-spoke LAN practicing EIGRP, GRE and router-based DHCP.

- **Build:** 23 devices: a hub router (HUB-RTR) and a spoke router (S6-RTR), three switches, three servers, IP phones, PCs, laptops and a printer.
- **Routing:** EIGRP AS 2704 on both routers, with a GRE tunnel between them.
- **DHCP:** five pools on the spoke router (two user, two voice, one printer).
- **Switching:** access switches with user VLAN 30 and trunks limited to VLANs 30, 40, 70, 90 and 999.
- **Management:** SSH v2 with a privileged local user, password encryption and vty lines limited to SSH.

Learning notes: a small network that let me focus on one skill at a time. The same tunnel address was reused on both ends, which I would now give unique addresses.

Lab 4: Project Network

Goal: a segmented network with an internal side, a firewall and a public-facing side with a honeypot.

- **Build:** 22 devices: HUB, OFFICE and PUB-RTR routers, a Layer 3 switch and access switches, an ASA 5506-X, two servers (MAIN and HONEYPOT), a cloud, PCs, laptops, IP phones and a printer.
- **Routing:** two EIGRP processes (AS 11 and AS 12) and GRE tunnels between the HUB, the office router and the Layer 3 switch.
- **VLANs:** separate VLANs for users (10, 11), the second tier (20, 21), 30, a server VLAN (50), native (90) and management (999).
- **Management:** a privileged local user, SSH on the vty lines and password encryption on the main devices.

Learning notes: this lab explores a public side and a decoy server, not only the internal network. The ASA and the public router are placed in the design but still on default configuration, so the security zones were designed here and implemented later.

Lab 5: Completed final

Goal: the smallest lab: two mirrored sites joined by a GRE tunnel, practicing EIGRP and basic device hardening.

- **Build:** 9 devices and 9 links: two 2911 routers, two 3560 Layer 3 switches, two 2960 switches and four laptops (Packet Tracer 7.2).
- **Sites:** each site has its own address block (172.168.0.0 and 172.16.0.0) with a /25 user subnet, a management /29 and point-to-point /30 links.
- **Routing:** EIGRP AS 5862 with a static default route redistributed, `ip routing` on the Layer 3 switches, and a GRE tunnel between the routers.
- **Layer 2:** access VLANs per site and trunks limited to specific VLANs including 999.
- **Hardening:** SSH v2 only on the vty lines (the unused lines are disabled), local login, password encryption, an enable secret, a login banner, and management IPs with a default gateway on the switches.

Learning notes: the smallest lab, and a clean example of the hardening habits (SSH only, management addressing, disabled unused lines). Both ends of the tunnel share one address, which I would now fix.

Progression and skills

Skill	Practiced in	Where it went
Subnetting and VLAN planning	Labs 3, 5	Three-Tier and CollapseCore address plans
EIGRP and OSPF with redistribution	Labs 1, 2	Three-Tier (OSPF, EIGRP, BGP), CollapseCore
eBGP	Lab 1	Three-Tier edge (AS 6550 to AS 6500)
GRE tunnels	Labs 1 to 5	Six tunnels, all validated, in CollapseCore
HSRP and EtherChannel	Lab 1, 2	Three-Tier HSRPv2; CollapseCore HSRP with tracking
Firewalling	Labs 2, 4 (ASA)	Zone-based firewall in CollapseCore
Hardening (SSH, passwords, management VLAN)	Lab 5	RADIUS AAA, port security, BPDU guard in CollapseCore

What this shows: the habit of learning a technology and then building a lab to prove it works, followed by redesigning the lab at a larger scale with more care about uniqueness of addresses, validation and security.

Next: combine these labs with the Three-Tier and CollapseCore overviews into one portfolio page, ordered from early practice to the hardened build.